



Computer Organization and Assembly Language

Lab # 2

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Task:

1. Write an instruction to:

- Move value 5 into AX
- Move AX into BX
- Add BX and AX

Answer:

MOV AX, 5 ; Move value 5 into AX

MOV BX, AX ; Move AX into BX

ADD AX, BX ; Add BX to AX ($AX = AX + BX$)

Explanation:

MOV AX, 5 $\rightarrow$ AX = 5

MOV BX, AX $\rightarrow$ BX = 5

ADD AX, BX $\rightarrow$ AX = 5 + 5 = 10

So, **AX = 10, BX = 5.**

2. Write output:

- MOV AX, 5
- MOV BX, 3
- ADD AX, BX
- SUB AX, 2

Answer:

MOV AX, 5 ; AX = 5

MOV BX, 3 ; BX = 3

ADD AX, BX ; AX = 5 + 3 = 8

SUB AX, 2 ; AX = 8 - 2 = 6

Final Output:

AX = 6

BX = 3

3. What are the differences between RAM and registers?

RAM	Registers
RAM stands for Random Access Memory	Registers are small storage locations inside the CPU
It stores programs and data currently in use	They store data and instructions that are being processed right now
Slower compared to registers	The fastest memory in the computer
Larger storage capacity	Very small storage capacity
Located outside the CPU	Located inside the CPU
Accessed indirectly by the CPU	Accessed directly by the CPU
Used for temporary storage during program execution	Used for immediate operations

Example:

RAM holds a program like a calculator app.

Registers hold values like AX and BX during calculations.
